Unraveling QTLs for Kernel Discoloration Resistance

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Malting barley (Hordeum vulgare L.) requires a grain protein content (GPC) of less than 13% to meet quality standards. Many breeding programs, including the North Dakota State University (NDSU) barley breeding program, have utilized a gene from the cultivar Karl, mapped to chromosome 6H, to achieve this reduction in GPC since the mid-1970s. Until recently, our low GPC breeding lines have exhibited unacceptable kernel discoloration, a critical quality parameter for maltsters who rely on visual assessments to avoid purchasing grain that may be affected by pathogens or weather. Using multiple bi-parental crosses, we could not determine whether the kernel discoloration and low GPC from Karl were due to linkage or pleiotropy. Recently, we identified breeding lines with low GPC and acceptable kernel color. To determine the genetic basis of this dissociation between the traits, a mapping panel of four cultivars and 201 breeding lines from NDSU were grown in three locations in North Dakota in 2023 and 2024. Kernel color was measured using a color spectrophotometer, while GPC was assessed using near-infrared reflectance. The mapping panel was genotyped with the 50K Illumina Infinium iSelect array. A genome-wide association study using Best Linear Unbiased Predictors revealed significant quantitative trait loci for both traits across multiple chromosomes. Notably, separate QTLs for low GPC and kernel discoloration were mapped to the centromeric region of chromosome 6H. Additionally, we identified QTLs unlinked to GPC, suggesting the potential to develop NDSU barley lines with optimal grain protein content and kernel discoloration resistance.

Introduction

- ✓ Kernel discoloration is associated with poor quality grain, affecting farmers gain —up to $1.00 per bushel in 2022.
- ✓ QTLs for discoloration have been identified to be co-located with the low-protein gene on chromosome 6H coming from the cultivar 'Karl'.
- ✓ 'Karl' has been used by many breeding programs to reduce the grain protein content to acceptable levels (i.e., <13.0%).

Figure 1: a) brightly colored barley kernels; b) differences in kernel discoloration across different barley varieties.

Materials and Methods

- ✓ 205 lines from the NDSU barley breeding program were evaluated in three locations and two years (2023-2024) in North Dakota using an unreplicated design with diagonal checks
- ✓ Kernel color measured via spectrophotometer (L-value) and grain protein assessed via NIR.

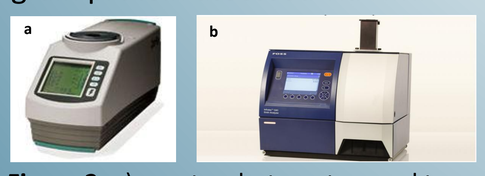

Figure 2. a) spectrophotometer used to measure kernel color; b) Near Infrared –NIR use to quantify grain protein.

- ✓ Best linear unbiased estimators - BLUEs were calculated for single locations.
- ✓ BLUEs were then used to calculate Best Linear Unbiased Predictors – BLUPs across locations analyses, respectively.
- ✓ Genotyping: 50K Illumina Infinium iSelect chip array.
- ✓ SNP data QAQC (MAF ≤ 0.05, missing data ≥ 0.10).
- ✓ GWAS used to identify significant QTLs.

	Plump	Grain Protein Content	Grain Color	Extract	S/T	Wort protein	Wort Clarity	Wort Color	Alpha-amylase	Diastatic Power	Beta-glucan	Free Amino Nitrogen
Plump	1.00000	-0.12128	-0.14254	0.20504	0.12269	0.06704	-0.01983	0.02743	0.02035	-0.06891	0.08144	0.05845
Grain Protein Content	0.00000	1.00000	-0.06399	-0.43179	-0.45038	0.22725	-0.01182	-0.03448	-0.13705	0.37635	0.04135	0.09857
Grain Color	0.00000	0.00110	1.00000	0.03591	0.05757	0.01062	0.06863	-0.02672	-0.03339	0.01119	-0.03318	0.02529
Extract	0.00000	0.00000	0.06721	1.00000	0.54575	0.26784	0.00530	0.11193	0.30451	-0.20982	-0.12075	0.27655
S/T	0.00000	0.00000	0.00333	0.00000	1.00000	0.70940	0.09079	0.29163	0.42389	-0.11892	-0.35152	0.60099
Wort Protein	0.00063	0.00000	0.58834	0.00000	0.00000	1.00000	0.08044	0.31517	0.36826	0.15939	-0.38136	0.77666
Wort Clarity	0.31234	0.54685	0.00046	0.78705	0.00000	0.00004	1.00000	0.25077	0.04453	0.02319	-0.11892	0.10115
Wort Color	0.16211	0.07884	0.17323	0.00000	0.00000	0.00000	0.00000	1.00000	0.18107	-0.06261	-0.19432	0.31912
Alpha-amylase	0.29970	0.00000	0.08877	0.00000	0.00000	0.00000	0.02318	0.00000	1.00000	0.13652	-0.23862	0.38925
Diastatic Power	0.00044	0.00000	0.56849	0.00000	0.00000	0.00000	0.23737	0.00141	0.00000	1.00000	-0.20660	0.09768
Beta-glucan	0.00003	0.03503	0.09076	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	1.00000	-0.40545
Free Amino Nitrogen	0.00287	0.00000	0.19736	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	1.00000

Figure 3: Pairwise correlation analysis for all quality traits tested for all the breeding lines evaluated from 2010-2023 in the NDSU Barley Breeding Program. Above the diagonal in red is the *r*-values between the traits and below is the *P*-value for the association between the traits

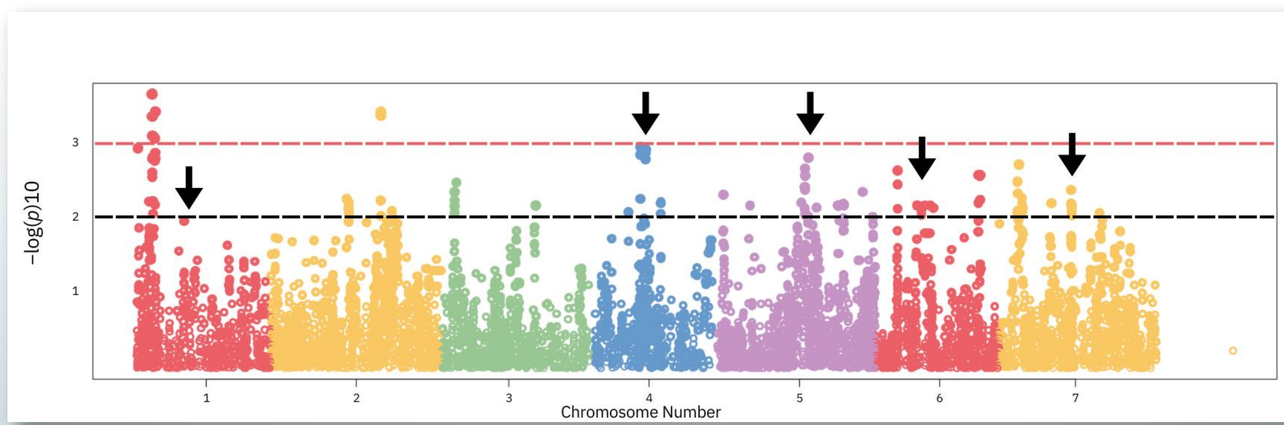

Figure 4: Manhattan plots indicating the SNPs that were identified for *kernel decolorization* in the GWAS study. Arrows indicate QTLs that are significantly associated in at least five of the six environments tested. The red and black dashed lines represent the thresholds for significance at the P≤0.001 and P≤0.01 significance levels.

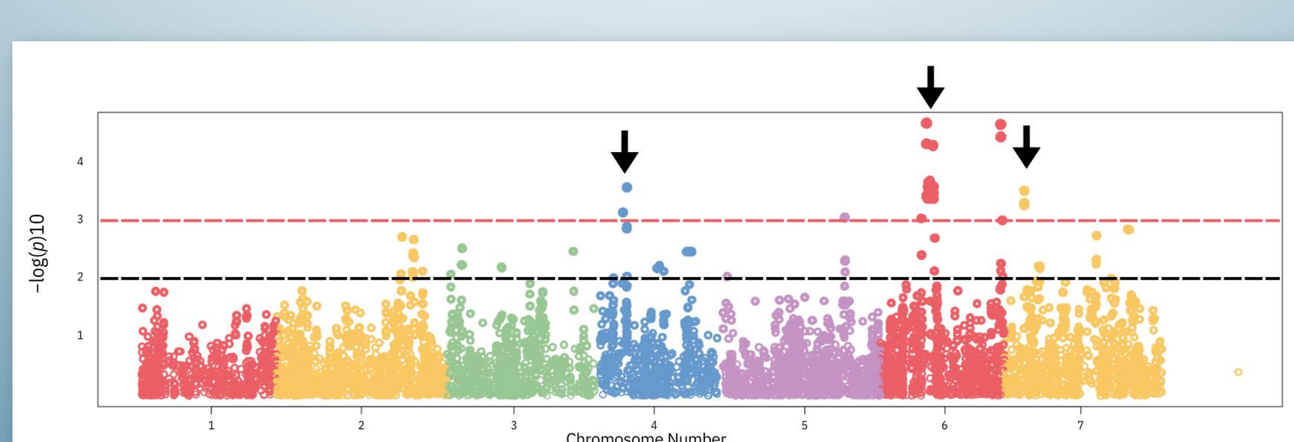

Figure 5: Manhattan plots indicating the SNPs that were identified for grain protein in the GWAS study. Arrows indicate QTLs that are significant in at least five of the six environments tested. The red and black dashed lines represent the thresholds for significance at the P≤0.001 and P≤0.01 significance levels.

Results and Discussion

- ✓ Significant QTLs for kernel color were identified in all chromosomes.
- ✓ Significant QTLs for grain protein content were identified in all chromosomes except 1H.
- ✓ Unsurprisingly, the significant QTLs for both kernel color and grain protein content in chromosome 6H were co-located in the centromeric region.
- ✓ The significant QTLs for the two traits in chromosomes 1H, 2H, 4H, 5H, and 6H (telomeric regions) do not appear to be co-located.
- ✓ Since several of the QTLs do not appear to be co-located, it indicates the potential to develop cultivars with acceptable grain protein content and kernel discoloration resistance.
- ✓ The large number of significant QTLs for kernel discoloration resistance suggests that the trait may be better suited for genomic selection model development than marker-assisted selection

Future Directions

- ✓ Collect more grain protein and kernel color data (2025) to determine if genetic haplotypes for each trait can be identified.
- ✓ If haplotypes can be identified for the chromosomes 4H and 6H protein QTL, assess whether lines carrying the low-protein haplotype on chromosome 4H and the high-protein haplotype on chromosome 6H exhibit improved kernel color.
- ✓ Develop a genomic selection model to select lines with ideal discoloration resistance and protein content.
- ✓ Develop a high-throughput model using an RGB camera that can predict L-values.

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